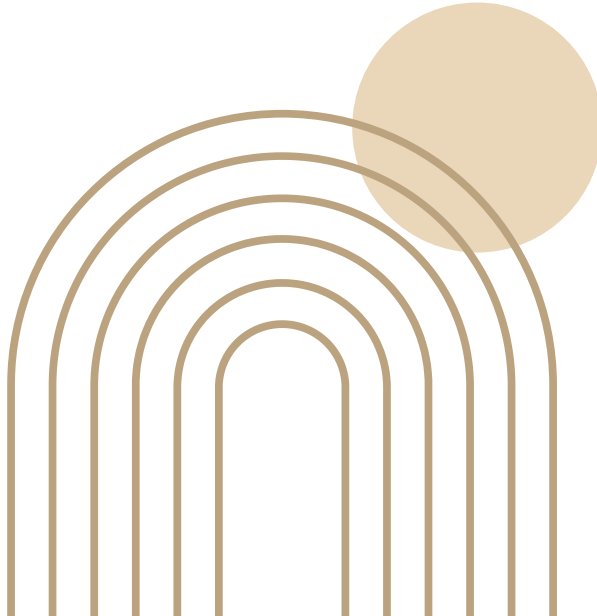




Agentic AI with LLMs to provide personalized educational assistance that would benefit the needs of every autistic student specifically



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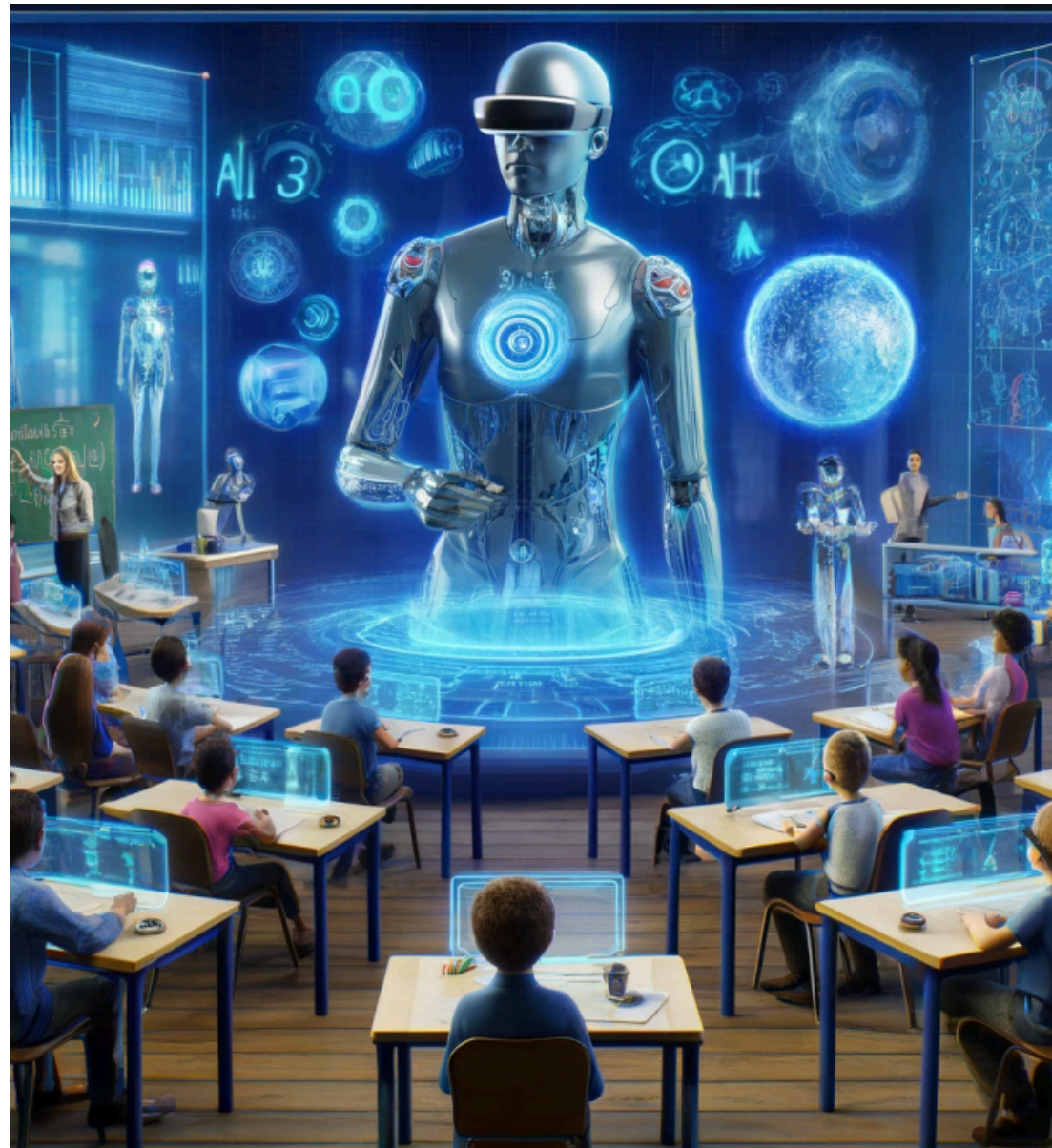


Hello!

My project, Agentic AI in Personalized Education Assistance to Autistic learners with the help of LLMs, seeks to understand how the AI can be customized to fit the specifications of autistic students. The system will have personalized support by using the learner profiles, the curriculum data, and the feedback. The idea is to develop an adaptive tutor to enhance involvement and the inclusiveness.



Problem Statement



This study addresses the gaps in current teaching methods for autistic learners, focusing on the need for more personalized and adaptive approaches.

Limitation of current teaching techniques

Novel educational technologies are typically generic and do not take into account the possibilities of the children with autism, which often implies a non-standard manner of encouraging learning.

Stress

Other consequences of the lack of personalization are the decrease in engagement and learning outcomes and stress among students and teachers.

Research Questions

Main Research Question

How can Agentic AI with LLMs to provide personalized educational assistance that would benefit the needs of every autistic student specifically?

Sub-Questions

- What kinds of data on learner profile (preferences, sensory sensitivities, communication styles) do we need in order to provide personalisation?
- What are the ways of modeling LLM-based AI to communicate in organized, streamlined and supportive manners befitting autistic learners?
- What can be done to use feedback loops (teacher/learner responses) to make the AI more adaptable as time goes on?



Research Design & Methods

Creating and Testing an Agentic AI in Supporting Personalised Learning to Autistic Students.



Data Collection

Use anonymized learner profiles, autism-specific instructional materials, dialogue datasets, and progress feedback.



Data Processing

creating documents, dividing them into chunks, converting them into vectors, and storing the vectors in a vector database through which efficient retrieval of the information is possible



AI Development

Use LLMs in combination with personalization techniques (prompt engineering, fine-tuning, reinforcement learning from feedback).



Evaluation

Evaluate on case studies / pilot datasets, taking measures on engagement, comprehension, and adaptability

Data Collection & Preprocessing

	Collection	Cleaning	Convert to Vectors	Vector Store DB
Week 1 – 2	Gather autism-relevant datasets:	Cleaning: Remove duplicates, anonymize sensitive data (student names, health info). create docs, split and then create chunks of docs.	Embedding the data sets into high dimensional vectoroes.	Store embeddings for efficient semantic search and retrieval.
Week 3 – 4	Australian National Autism Strategy Resources, Research Papers on Autistic Learning	Standardizing: Convert into uniform format (CSV, JSON, or text), tokenize.	Hugging Face Embeddings. OpenAi Embedding	FAISS CHROMA PINECONE

Data Analysis

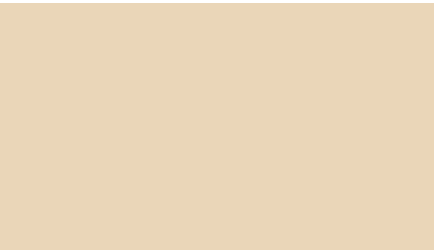
Websites contribute about 40,000 tokens with practical guidelines and teaching strategies. Research papers add 250,000 tokens covering learner profiles, teaching methods, and assessments. Most tokens come from research papers, providing evidence-based insights, while websites offer scenario-based guidance. The combined dataset supports LLM training for personalized autistic learning.

- Removed duplicates, references and sensitive data
- Chunked larger texts into 3k token to feed LLM
- **Used Sentence-Transformers (all-MiniLM-L6-v2) Embedding Dimension: 384**

Limitations

- Lack of real-time interaction data for some students
- Multimodal inputs (images, video) may need separate embedding





Building the LLM Framework

- I am using Large Language Model (LLM) as the foundation.
 - RAG and prompt-engineering the LLM with autism-specific teaching material.
 - Design the agent to adapt lessons and communication styles to individual learner needs.
 - Include multimodal support (text, visuals, audio) for accessibility.
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Feedback Loops & Adaptation

- Collect feedback from simulated learners, teachers, or caregivers.
- Use RLHF or preference-based tuning to refine AI responses iteratively.





Goal

Create an adaptive, inclusive AI tutor that improves engagement, comprehension, and personalized learning outcomes for autistic learners.



THANK YOU

